

Zakira Farheen

Senior DevOps · SRE · DevSecOps · MLOps Engineer

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SUMMARY

Senior DevOps, SRE, and DevSecOps engineer with 15+ years making infrastructure that is fast to deploy, hard to break, and secure by design. I build and run cloud-native platforms and CI/CD for demanding, regulated environments (Aviva, Lloyds Bank), and I take machine-learning models from research into dependable production with MLflow, Kubeflow, and SageMaker. My focus now is LLM, RAG, and agentic systems that are secure and auditable, not just impressive in a demo. If a team needs models that ship reliably and stay healthy in production, that is the problem I solve.

CORE SKILLS

Cloud: AWS, Azure, GCP

Infrastructure as code: Terraform, policy-as-code

CI/CD: Jenkins, Azure DevOps

Containers: Docker, Kubernetes

Practices: Chaos engineering, observability, monitoring & alerting

MLOps: MLflow, Kubeflow, SageMaker, Weights & Biases

DevSecOps: SAST, container scanning, secrets management

Reliability/SRE: SLOs, error budgets, incident response, RCA

Generative AI: LLMs, RAG, LangGraph

Domains: Financial services, regulated/compliance environments

EXPERIENCE

Principal MLOps Engineer

Feb 2026 – Present

Cyber Resiliency Labs · Ontario, Canada

- Lead the MLOps practice, helping businesses and non-profits get machine-learning models into production and keep them reliable, secure, and cost-effective.
- Design and operate end-to-end ML pipelines and the cloud infrastructure underneath, building on AWS, Terraform, MLflow, Kubeflow, and SageMaker.
- Advise teams on production ML and generative-AI systems (LLMs and RAG) that are secure and auditable, not just functional.

Senior DevOps Engineer

Jan 2024 – Present

Aviva Canada (via Tata Consultancy Services) · Toronto, ON

- Keep the Guidewire platform dependable on AWS, using Terraform and automation to make releases routine and recovery fast.
- Built the monitoring and alerting that surfaces production issues before they reach customers, cutting unplanned downtime.
- Own root-cause analysis and incident response, turning outages into fixes that stop the same problem recurring.

Lead DevOps / Cloud Consultant

May 2021 – Jan 2024

Aviva UK (via Tata Consultancy Services) · United Kingdom

- Led Aviva UK's DevOps and cloud transformation, replacing manual releases with fully automated pipelines that ship faster and break less.
- Standardized infrastructure on Azure with Terraform and brought CI/CD discipline across delivery teams through Jenkins and Azure DevOps.

- Ran cross-cloud workloads on AWS and coached teams to invest in pipeline quality, which paid back in fewer late-stage failures.

Senior DevOps Engineer

Jan 2019 – May 2021

Lloyds Bank, UK (via Tata Consultancy Services)

- Rebuilt the CI/CD pipeline for a suite of .NET applications at one of the UK's largest banks, moving delivery onto AWS and GCP with full automation.
- Replaced slow, manual releases with automated ones, so the team shipped faster and with far more confidence.
- Held delivery speed and strict security, compliance, and audit controls in balance, a non-negotiable in banking.

Senior DevOps Engineer

Apr 2017 – Jan 2019

Lloyds Bank, UK (via Teamware Solutions, Quantum Leap Consulting)

- Built CI/CD pipelines for .NET applications and migrated workloads to AWS and GCP.
- Took manual steps out of deployment and made pipelines more reliable, while keeping everything auditable under strict change control.

Technology Analyst

Aug 2010 – Aug 2015

Nordstrom, US (via Infosys)

- Owned release engineering and build management for a large retail environment: artifact versioning, builds, and deployments to on-premise infrastructure.
- Learned the full path from commit to production end to end, the foundation for the automation work that followed.

Research Fellow

Sep 2008 – Aug 2010

Osmania University

- Applied research on large datasets and structured data analysis, the start of a path that now runs through production ML and generative AI.

EDUCATION

M.Eng, Embedded Systems & VLSI Design

Osmania University · 2008 – 2010

B.Tech, Electronics & Communication Engineering

Jawaharlal Nehru Technological University · 2004 – 2008

CERTIFICATIONS

- HashiCorp Certified: Terraform Associate
- AWS Certified Cloud Practitioner
- Microsoft Certified: DevOps Engineer (AZ-400)
- Professional Scrum Master I
- IBM Certified Solution Advisor: Cloud Computing Architecture V5

PUBLICATION

Content-Based Image Retrieval using Independent Component Analysis.

References available on request. · crlab.ca/MLOps